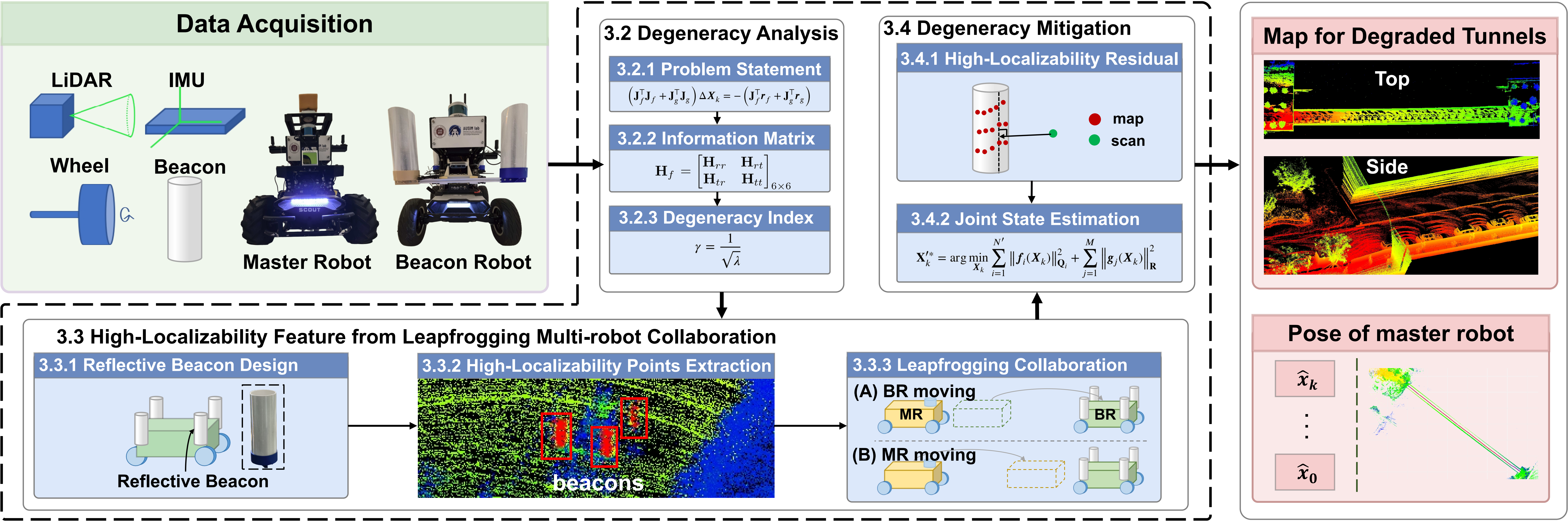

Data Acquisition
LiDAR
IMU
Wheel
Beacon
Master Robot
Beacon Robot
3.2 Degeneracy Analysis
3.2.1 Problem Statement
$\left(\mathbf{J}_f^{\top}\mathbf{J}_f + \mathbf{J}_g^{\top}\mathbf{J}_g\right)\Delta X_k = -\left(\mathbf{J}_f^{\top} r_f + \mathbf{J}_g^{\top} r_g\right)$
3.2.2 Information Matrix
$\mathbf{H}_f = \begin{bmatrix} \mathbf{H}_{rr} & \mathbf{H}_{rt} \\ \mathbf{H}_{tr} & \mathbf{H}_{tt} \end{bmatrix}_{6\times 6}$
3.2.3 Degeneracy Index
$\gamma = \frac{1}{\sqrt{\lambda}}$
3.4 Degeneracy Mitigation
3.4.1 High-Localizability Residual
map
scan
3.4.2 Joint State Estimation
$\mathbf{X}_k'^{*} = \arg\min_{\mathbf{X}_k} \sum_{i=1}^{N'} \left\| f_i(\mathbf{X}_k) \right\|_{\mathbf{Q}_i}^2 + \sum_{j=1}^{M} \left\| g_j(\mathbf{X}_k) \right\|_{\mathbf{R}}^2$
Map for Degraded Tunnels
Top
Side
Pose of master robot
$\hat{x}_k$
$\hat{x}_0$
3.3 High-Localizability Feature from Leapfrogging Multi-robot Collaboration
3.3.1 Reflective Beacon Design
Reflective Beacon
3.3.2 High-Localizability Points Extraction
beacons
3.3.3 Leapfrogging Collaboration
(A) BR moving
MR
BR
(B) MR moving